

HALIFAX NORTHAMPTON, NC, USA

WMO: 720649

Lat: 36.330N

Lon: 77.635W

Elev: 44

StdP: 100.80

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00231

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.2	-17.1	0.8	-3.8	-13.9	1.1	-2.6	9.5	12.3	8.5	10.8	2.1	240	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	35.0	24.6	32.9	24.4	32.2	24.3	26.8	31.4	25.8	30.7	25.1	30.1	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	28.8	24.8	19.9	28.4	23.7	18.6	27.2	84.1	31.6	79.9	30.6	76.5	29.8	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.2	7.2	5.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-17.6	36.8	9.5	2.3	-24.4	38.4	-30.0	39.8	-35.4	41.0	-42.3	42.7
			-18.0	27.4	9.3	1.0	-24.6	28.1	-30.0	28.7	-35.2	29.2	-42.0	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	4.3	6.4	9.6	15.1	20.4	24.1	26.1	24.7	22.0	15.7	9.6	6.6	(6)	
(7)		DBStd	8.76	5.49	5.47	5.50	4.78	3.98	3.02	2.59	2.34	2.98	4.31	4.62	5.19	(7)	
(8)		HDD10.0	608	190	129	79	11	0	0	0	0	0	5	61	133	(8)	
(9)		HDD18.3	1932	436	337	276	117	27	1	0	0	7	104	263	365	(9)	
(10)		CDD10.0	2591	13	27	66	165	324	422	499	456	361	183	49	27	(10)	
(11)		CDD18.3	874	0	2	3	21	92	173	241	198	117	24	2	2	(11)	
(12)		CDH23.3	7832	0	11	45	202	819	1671	2393	1641	847	192	7	6	(12)	
(13)		CDH26.7	3097	0	1	6	35	279	696	1125	642	275	37	0	1	(13)	
(14)	Wind	WSAvg	2.2	2.4	2.6	2.7	2.9	2.2	2.1	1.9	1.5	1.7	2.1	2.1	2.1	(14)	
(15)	Precipitation	PrecAvg	1101	84	64	107	77	84	104	122	111	111	83	77	83	(15)	
(16)		PrecMax	1434	156	137	206	188	150	261	206	273	479	172	206	154	(16)	
(17)		PrecMin	868	29	13	23	30	22	39	52	16	3	0	14	22	(17)	
(18)		PrecStd	160	34	30	43	36	37	62	41	58	97	43	47	32	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	25.0	27.5	30.0	33.0	36.2	37.6	35.0	33.7	30.0	24.0	22.9	(19)	
(20)		MCWB	16.7	17.9	17.8	19.1	23.0	23.6	24.5	24.7	22.1	22.8	17.7	19.1	19.1	(20)	
(21)		2%	DB	17.9	22.1	24.7	27.4	31.4	33.8	35.8	32.8	31.2	27.4	22.2	20.9	(21)	
(22)		MCWB	14.2	17.4	16.9	19.2	22.2	23.7	25.1	24.5	23.1	20.8	17.1	18.3	18.3	(22)	
(23)		5%	DB	16.2	19.1	22.2	26.0	30.0	32.3	33.1	32.1	29.9	25.8	20.1	17.7	(23)	
(24)		MCWB	13.0	15.2	15.4	18.3	21.9	23.9	25.0	24.5	22.6	20.1	16.2	15.4	15.4	(24)	
(25)		10%	DB	13.0	17.0	19.9	23.7	27.7	31.0	32.3	30.2	27.8	23.0	17.7	15.9	(25)	
(26)		MCWB	10.2	13.3	14.1	17.1	21.1	23.3	24.9	24.0	22.1	19.0	14.4	13.7	13.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	19.7	20.8	22.6	24.7	26.2	27.9	27.4	26.0	25.2	19.8	20.7	(27)	
(28)		MDCB	20.7	22.9	24.1	26.1	30.8	30.7	32.6	31.4	28.5	28.5	21.6	22.2	22.2	(28)	
(29)		2%	WB	16.2	18.0	18.4	21.1	23.6	25.2	27.1	26.2	24.6	23.2	18.6	18.2	(29)	
(30)		MDCB	17.2	20.4	23.1	24.9	29.0	31.1	32.0	30.5	28.4	24.8	21.1	19.9	19.9	(30)	
(31)		5%	WB	13.7	16.4	17.1	19.9	22.9	24.7	26.3	25.4	24.0	21.8	17.1	16.3	(31)	
(32)		MDCB	15.6	18.4	20.0	24.0	27.8	30.3	31.5	29.4	28.0	24.3	18.9	17.5	17.5	(32)	
(33)		10%	WB	11.0	13.0	15.5	18.5	22.1	24.0	25.5	24.6	23.1	20.1	15.3	13.6	(33)	
(34)		MDCB	13.0	15.9	18.7	22.4	26.3	29.4	30.9	28.8	26.4	22.4	17.7	15.3	15.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.7	11.3	12.3	13.2	11.8	11.8	11.4	10.4	10.5	12.1	12.1	10.7	10.7	(35)	
(36)		5% DB	MDCDBR	14.4	14.7	15.5	14.3	13.9	13.1	12.7	11.9	12.9	13.6	13.8	12.8	12.8	(36)
(37)		MCWBR	10.7	9.8	8.7	7.3	5.8	4.7	4.6	4.4	5.4	7.5	8.9	9.9	9.9	(37)	
(38)		5% WB	MDCDBR	12.6	14.1	12.5	11.8	11.3	11.5	11.6	10.3	10.3	10.1	10.7	11.5	11.5	(38)
(39)		MCWBR	11.5	10.8	8.6	7.0	5.3	5.1	5.4	4.9	5.3	6.3	8.3	9.9	9.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.319	0.349	0.394	0.442	0.492	0.520	0.518	0.407	0.360	0.322	0.306	0.306	(40)	
(41)		taud	2.521	2.478	2.427	2.308	2.201	2.110	2.048	2.042	2.350	2.466	2.527	2.555	2.555	(41)	
(42)		Ebn at Noon	897	921	918	889	847	802	775	768	847	862	867	872	872	(42)	
(43)		Edh at Noon	80	93	108	128	145	158	167	165	115	94	79	73	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.48	3.19	4.18	5.32	5.92	6.38	6.04	5.47	4.48	3.60	2.74	2.18	2.18	(44)	
(45)		RadStd	0.19	0.37	0.32	0.41	0.46	0.41	0.30	0.30	0.38	0.45	0.22	0.18	0.18	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (69 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	+151	+105	(47)

Nomenclature: See separate page